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COMMON QUARTERLY EXAMINATION - 2025 Reg. No. | | | | | | |

XII - MATHEMATICS

Time Allowed : 3.00 Hrs. *Maximum Marks: 90*

Part - I

I. Choose the correct answer: **20 x 1 = 20**

1. If $A=\begin{bmatrix}3 & 5\\ 1 & 2\end{bmatrix}$, B = adj A and C = 3 A then $\frac{|adjB|}{|c|}=$

   a) $\frac{1}{3}$ b) $\frac{1}{9}$ c) $\frac{1}{4}$ d) 1

2. If $P=\begin{bmatrix}1 & x & 0\\ 1 & 3 & 0\\ 2 & 4 & -2\end{bmatrix}$ is the adjoint of 3 x 3 matri x A and $|A|=4$, then x is

   a) 15 b) 12 c) 14 d) 11

3. If $P(A)=P([A|B])$, then the system AX = B of linear equation is

   a) Consistent and has a unique solution b) consistent
   c) Consistent and has infinitely many solutions d) inconsistent

4. If $Z=\frac{(\sqrt{3}+1)^3(3i+4)^2}{(8+6i)^2}$ then $|Z|$ is equal to

   a) 0 b) 1 c) 2 d) 3

5. The principal argument of $\frac{3}{-1+i}$ is

   a) $\frac{-5\pi}{6}$ b) $\frac{-2\pi}{3}$ c) $\frac{-3\pi}{4}$ d) $\frac{-\pi}{2}$

6. If $\omega\neq 1$ is a cubic root of unity and $\begin{vmatrix}1 & 1 & 1\\ 1 & -\omega^2-1 & \omega^2\\ 1 & \omega^2 & \omega^7\end{vmatrix}=3k$, then k is equal to

   a) 1 b) -1 c) $\sqrt{3i}$ d) $-\sqrt{3i}$

7. If f and g are polynomials of degree m and n respectively and if $h(x)=(f\circ g)(x)$, then the degree of h is

   a) mn b) m + n c) m^n d) n^m

8. The polynomial x^3-kx^2+9x has three real zeros if and only if, k satisfies

   a) $|k|\leq 6$ b) $k=0$ c) $|k|>6$ d) $|k|\geq 6$

9. The number of real numbers in $[0,2\pi]$ satisfying $\sin^4 x-2\sin^2 x+1$ is

   a) 2 b) 4 c) 1 d) ∞

10. $\sin^{-1}x=2\sin^{-1}\alpha$ has a solution, then

    a) $|\alpha|\leq\frac{1}{\sqrt{2}}$ b) $|\alpha|\geq\frac{1}{\sqrt{2}}$ c) $|\alpha|<\frac{1}{\sqrt{2}}$ d) $|\alpha|>\frac{1}{\sqrt{2}}$

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11. The domain of the function defined by $f(x)\sin^{-1}\sqrt{x-1}$ is

a) [1,2] b) [-1,1] c) [0,1] d) [-1,0]

12. $\text{Sin}^{-1}\left(\tan\frac{\pi}{4}\right)-\sin^{-1}\left(\sqrt{\frac{3}{X}}\right)=\frac{\pi}{6}$. Then x is a root of the equation.

a) $x^2 - x - 6 = 0$ b) $x^2 - -x - 12 = 0$ c) $x^2 + x - 12 = 0$ d) $x^2 + x - 6 = 0$

13. The equation of the normal to the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ which is parallel to the line $2x + 4y = 3$

a) $x + 2y = 3$ b) $x + 2y + 3 = 0$ c) $x + 4y + 3 = 0$ d) $x - 2y + 3 = 0$

14. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

a) Zab b) ab c) $\sqrt{ab}$ d) $\frac{a}{b}$

15. If the coordinates at one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are (11,2), the coordinate of the other end is

a) (-5, 2) b) (2, -5) c) (5, -2) d) (-2, 5)

16. The volume of the parallelepiped with its edge represented by the vectors

$\hat{i}+\hat{j},\ \hat{i}+2j,\ \hat{i}+\hat{j}+\pi\hat{k}$ is

a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) π d) $\frac{\pi}{4}$

17. If $\vec{a},\vec{b},\vec{c}$ are three non - coplanar vectors such that $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\frac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$, and $\vec{b}$ is

a) $\frac{\pi}{2}$ b) $\frac{3\pi}{4}$ c) $\frac{\pi}{4}$ d) π

18. If the direction cosines of a line are $\frac{1}{c},\frac{1}{c},\frac{1}{c}$ then

a) $c = \pm 3$ b) $c = \pm\sqrt{3}$ c) $c>0$ d) $0<c<1$

19. If ρ represents the rank and A and B are n x n matrices then

a) $\rho(A+B) = \rho(A) + \rho(B)$ b) $\rho(AB) = \rho(A).\ \rho(B)$

c) $\rho(A-B) = \rho(A) - \rho(B)$ d) $\rho(A+B) \leq n$

20. The number of tangents to the circle from inside the circle is

a) 2 real b) 0 c) 2 imaginary d) can't be determined

Part - II

II. Answer any 7 questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. Find the inverse (if it exists) of the following : $\begin{bmatrix}-2 & 4\\ 1 & -3\end{bmatrix}$

22. Show that $\left(2+i\sqrt{3}\right)^{10}+\left(2-i\sqrt{3}\right)^{10}$ is real

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23. Find the principal argument Argz, when $z = \frac{-2}{1+i\sqrt{3}}$

24. If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum \frac{1}{\beta\gamma}$ in terms of the coefficients.

25. Find the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

26. Find the domain of $\tan^{-1}\left(\sqrt{9-x^2}\right)$.

27. Obtain the equation of the circle for which (3,4) and (2,-7) are the ends of a dimeter.

28. The orbit of Halley's Comet is an ellipse 36.18 astronomical units long and by 9.12 astronomical units wide. Find its eccentricity.

29. Determine whether the three vectors $2\hat{i}+3\hat{j}+\hat{k}$, $\hat{i}-2\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}+3\hat{k}$ are coplanar.

30. Find the distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) **7 x 3 = 21**

31. Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$ by reducing it to a row-echelon form.

32. Solve the following system of linear equations by matrix inversion method
$2x + 5y = -2$, $x + 2y = -3$.

33. If $z = x + iy$ is a complex number such that $\left|\frac{z-4i}{z+4i}\right| = 1$, show that the locus of z is real axis.

34. Solve the equation $x^3 - 5x^2 - 4x + 20 = 0$

35. Find the domain of $\cos^{-1}\left(\frac{2+\sin x}{3}\right)$

36. Prove that $\tan^{-1}\frac{2}{11} + \tan^{-1}\frac{7}{24} = \tan^{-1}\frac{1}{2}$

37. Find the length of Latus rectum of the parabola $y^2 = 4ax$.

38. If $\vec{a} = 2\hat{i}+3\hat{j}-\hat{k}$, $\vec{b} = -\hat{i}+2\hat{j}-4\hat{k}$, $\vec{c} = \hat{i}+\hat{j}+\hat{k}$, then find the value of $(\vec{a} \times \vec{b}).(\vec{a} \times \vec{c})$

39. If $\vec{a}, \vec{b}, \vec{c}$ are three non - coplanar vectors, prove that $[\vec{a}+\vec{c}, \vec{a}+\vec{b}, \vec{a}+\vec{b}+\vec{c}] = [\vec{a}, \vec{b}, \vec{c}]$.

40. Find the area and circumference of the circle $x^2 + y^2 - 2x + 5y + 7 = 0$.

Part - IV

IV. Answer all the questions. **7 x 5 = 35**

41. a) Solve the following system of liner equations by Cramer's rule:
$3x + 3y - z = 11$, $2x - y + 2z = 9$, $4x + 3y + 2z = 25$

(OR)

b) If $\omega \neq 1$ is a cube root of unity, show that the roots of the equation $(z-1)^3 + 8 = 0$ are -1, $1-2\omega$, $1-2\omega^2$

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42. a) If the equation $x^2+px+q=0$ and $x^2+p'x+q'=0$ have a common root, show it must be equal to $\frac{pq'-p'q}{q-q'}$ or $\frac{q-q'}{p'-p}$

(OR)

b) Prove that $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$.

43. a) Find the equations of tangent and normal to the ellipse $x^2+4y^2=32$ when $\theta=\frac{\pi}{4}$.

(OR)

b) Using vector method, prove that $\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$.

44. a) Investigate the values of λ and μ the system of linear equations $2x+3y+5z=9$, $7x+3y-5z=8$, $2x+3y+\lambda z=\mu$, have
(i) no solution (ii) a unique solution (iii) an infinite number of solutions.

(OR)

b) Find all zeros of the polynomial $x^6-3x^5-5x^4+22x^3-39x^2-39x+135$, if it is known that $1+2i$ and $\sqrt{3}$ are two of its zeros.

45. a) Let z_1, z_2 and z_3 be complex numbers such that $|z_1|=|z_2|=|z_3|=r>0$ and $z_1+z_2+z_3\neq 0$. prove that $\left|\frac{z_1z_2+z_2z_3+z_3z_1}{z_1+z_2+z_3}\right|=r$.

(OR)

b) Find the foci, vertices and length of major and minor axis of the conic $4x^2+36y^2+40x-288y+532=0$.

46. a) Find the value of $\tan^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$. *(OR)*

b) If $A=\begin{bmatrix}-4 & 4 & 4\\ -7 & 1 & 3\\ 5 & -3 & -1\end{bmatrix}$ and $B=\begin{bmatrix}1 & -1 & 1\\ 1 & -2 & -2\\ 2 & 1 & 3\end{bmatrix}$ find the products AB and BA and hence solve the system of equations $x-y+z=4$, $x-2y-2z=9$, $2x+y+3z=1$.

47. a) Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point (2,3,6) and parallel to the straight lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-3}{1}$ and $\frac{x+3}{2}=\frac{y-3}{-5}=\frac{z+1}{-3}$ **(OR)**

b) Two coast guard stations are located 600 km apart at points A(0,0) and B(0,600). A distress singnal from a ship at p is received at slightly different times by two stations. It is determined that the ship is 200 km farther from station A than it is from station B. Determine the equation of hyperbola that passes through the location of the ship.

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